

ResearchMate

LLM Powered Academic Research Planning and
Decision Support Agent

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Individual Project Presentation

Problem Statement

The Challenge

- Students begin with broad research ideas.
- Research aims and objectives are often poorly aligned.
- Research questions can lack focus.
- Literature searches become inefficient.
- General AI tools do not provide a structured planning workflow.

ResearchMate Solution

- ✓ Validates the research topic
- ✓ Creates a structured research plan
- ✓ Generates aims, objectives and questions
- ✓ Suggests literature-search keywords
- ✓ Produces a structured academic report

Key Takeaway: ResearchMate transforms an initial research idea into a structured and coherent academic research plan.

Project Aim & Objectives

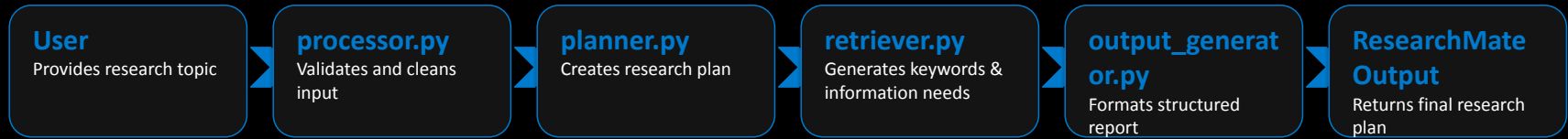
Project Aim

Design and implement an intelligent academic research planning agent that transforms a user's research idea into a structured research plan.

Project Objectives

- ✓ Validate and clean the user research topic.
- ✓ Generate a structured academic research plan.
- ✓ Produce aligned research aims, objectives and questions.
- ✓ Suggest literature-search keywords and information requirements.
- ✓ Generate a clear, structured research planning report.
- ✓ Demonstrate a modular, testable intelligent agent architecture.

ResearchMate Intelligent Agent Workflow



Key Design Principle: Each module performs a single responsibility, making the agent modular, maintainable, testable and easy to extend.

Implementation & Code Structure

Project Modules

- main.py — Application entry point
- agent.py — Coordinates the workflow
- processor.py — Validates and cleans input
- planner.py — Generates research plans
- retriever.py — Creates keywords & information needs
- output_generator.py — Builds the final report
- tests/ — Unit & functional testing

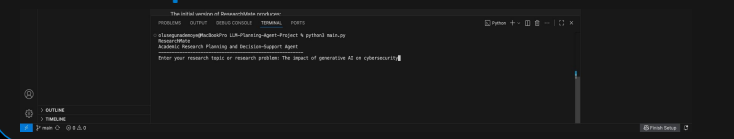
Key Design Decisions

- ✓ Modular architecture
- ✓ Single Responsibility Principle
- ✓ Separation of concerns
- ✓ Deterministic planning for predictable testing
- ✓ Reusable components
- ✓ Extensible for future LLM integration

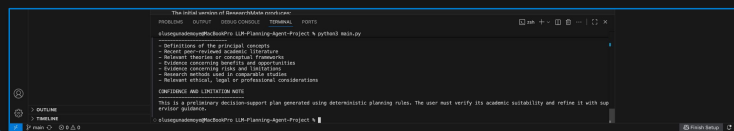
The modular design enables each component to be developed, tested and replaced independently.

ResearchMate in Action

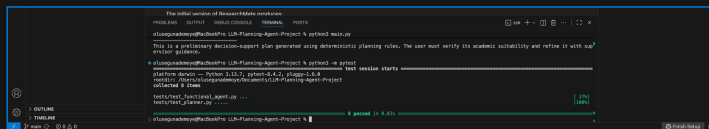
1. User Input



2. Generated Research Plan



3. Automated Testing



Intelligent Agent Execution

- ✓ User research topic received
- ✓ Input validated successfully
- ✓ Research plan generated
- ✓ Literature-search keywords produced
- ✓ Structured report created
- ✓ Automated tests completed successfully (8 passed)

This demonstration confirms that ResearchMate successfully performs the complete intelligent-agent workflow from user input through to structured academic output.

Testing & Remediation

Testing Activities

- Unit Testing: Verified individual planner and processor functions.
- Functional Testing: Validated the complete end-to-end workflow.
- Input Validation: Rejected empty or invalid research topics.
- Regression Testing: Confirmed improvements did not break existing features.

Current Results

- ✓ 8 Automated Tests Passed
- ✓ No Failing Tests
- ✓ All current automated tests passing

Remediation Examples

- Updated failing functional test after improving report headings.
- Corrected planner test discovery and file naming.
- Added input validation to improve robustness.

Evaluation, Limitations & Ethical Considerations

Strengths

- ✓ Modular and maintainable architecture
- ✓ Structured end-to-end workflow
- ✓ Automated testing with 8 passing tests
- ✓ Clear and consistent research planning output

Limitations

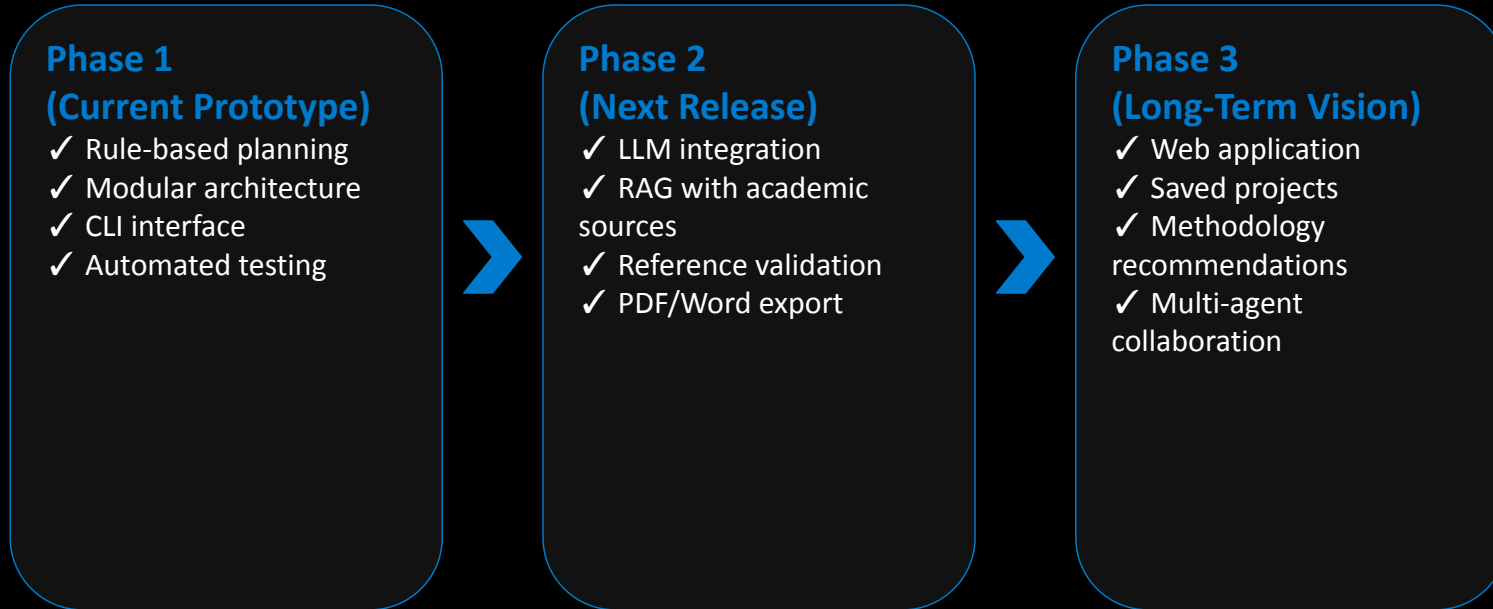
- Rule-based planner (no live LLM)
- No live academic database retrieval
- Command-line interface only
- Outputs require human academic review

Ethical Considerations

- Supports planning rather than replacing academic judgement
- Users must verify outputs and references
- Helps reduce over-reliance on generative AI
- Designed to promote responsible AI use

ResearchMate is intended as an academic decision-support tool, not a replacement for independent research or supervisor guidance.

Future Improvements & Product Roadmap



Roadmap: Evolve ResearchMate from a deterministic planning tool into an AI-powered academic research assistant.

Conclusion & Key Takeaways

Project Achievements

- ✓ Designed and implemented a modular intelligent planning agent.
- ✓ Developed a complete end-to-end planning workflow.
- ✓ Produced structured academic research plans.
- ✓ Created automated unit and functional tests (8 passed).
- ✓ Documented architecture, design decisions and evaluation.

Key Takeaways

- Planning before generation improves structure and consistency.
- Modular architecture supports maintainability and future enhancement.
- Testing and documentation strengthen software quality.
- ResearchMate is a foundation for future LLM-powered academic assistance.

Thank you